

SARAN JAYA THILAK

Data Engineer | Cloud Infrastructure | CI/CD Automation | Machine Learning Proficiency

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SUMMARY

Data science professional with a strong background in machine learning, cloud infrastructure, and automation. Experienced in designing scalable architectures on AWS and GCP, and developing end-to-end CI/CD pipelines to streamline deployment workflows. Passionate about applying data science and AI to solve real-world problems, with a focus on building reliable, impactful, and scalable solutions.

KEY ACHIEVEMENTS

- Deployment Cycle Improvement

Improved deployment cycle by 25% using CI/CD automation pipelines.
- Inventory Cost Reduction

Reduced inventory costs by 15% through efficient inventory management strategies.

EDUCATION

- 04/2025 - 06/2025

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Boot Camp, Data science and AI
Le Wagon
- 04/2022 - 03/2023

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Master's degree, Data Science
University of Europe for Applied Sciences
- 01/2013 - 12/2014

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PG DIPLOMA; NIELIT, VLSI AND EMBEDDED SYSTEMS HW DESIGNING
Delhi University
- 01/2009 - 12/2013

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Bachelor of Technology (BTech), Electrical, Electronics and Communications Engineering
Cochin University of Science and Technology

EXPERIENCE

- 07/2023 - 02/2025

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Production Associate
Berlin, Germany
Tesla

Improved assembly line efficiency by {15%} by maintaining optimal inventory for {24/7} production demands.

Collaborated cross-functionally with production, engineering, and logistics teams to streamline workflows and ensure operational efficiency.

Partnered with engineering and quality teams to identify bottlenecks and recommend process improvements, enhancing overall production throughput.
- 06/2019 - 02/2022

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Cloud Support Engineer
Bengaluru, India
Huawei

Designed and deployed scalable, cloud-native infrastructures on AWS, using Terraform to automate provisioning and enhance infrastructure consistency.

Built CI/CD pipelines that accelerated deployment cycles and reduced manual configuration errors.

Specialized in AWS services including EC2, S3, Lambda, and DynamoDB while implementing secure and compliant architectures.

Improved system reliability and cost efficiency through proactive monitoring and resource optimization.
- 05/2016 - 05/2019

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Operations and Maintenance Engineer
Bengaluru, India
Huawei

Led a team within the Global Network Operations Center (GNOC), managing 24/7 monitoring and incident escalation for MTN's multinational telecom infrastructure.

Mentored and trained junior engineers, driving knowledge sharing and technical upskilling across the team.

Oversaw operations across IP, transmission, mobile core, and RAN domains, maintaining high network availability and SLA compliance.
- 08/2014 - 05/2016

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Operations and Maintenance Engineer
Bangalore, India
Nokia

Maintained telecom infrastructure across Karnataka, ensuring 99.99% network availability and quick resolution of faults.

Reduced downtime through proactive system monitoring, timely fault management, and coordination with cross-functional teams.

Led preventive and corrective maintenance for key network components including BTS, BSC, and transmission systems, improving long-term stability.

SKILLS

Python	APIs & Web Scraping		TensorFlow		Big Data		Git	GitLab	AB Testing		Keras	CNN	
Natural Language Processing			sentiment Analysis			Machine Learning			SQL	PostgreSQL		Google Cloud Platform	
Google BigQuery		DBT	Apache Airflow		DAX	Terraform		CI/CD	Microsoft Power BI			Tableau	Linux
Deep Learning		IAM	RDS	VPC	LSTM	K8s	Docker	PyTorch	MLOps	MLflow	scikit-Learn		
LLM	Langchain	Flask	NumPy										

PROJECTS

AI Knowledge Assistant (Local RAG + LangChain)

Developed an AI chat interface to interact with PDF documents by integrating Google Drive, Notion, Google Sheets, and SQL databases. Utilized **LangChain** for retrieval-augmented generation (**RAG**) and orchestration of local large language models (**LLMs**).

Divvy Bikes End to End Data Pipeline

Built an end-to-end data pipeline on **GCP** using **Terraform** for infrastructure setup, **Airflow** for orchestration, and **DBT** for SQL-based data transformation. Loaded data into **BigQuery** and visualized insights through **Google Data Studio**, enabling scalable, automated processing of the Divvy Bikes dataset.

PUBLICATIONS

Performance Ratio Estimation of Solar Power Plants Using Machine Learning Algorithms

IEEE
📅 09/2023 🔗 <https://ieeexplore.ieee.org/document/10317646>
Applied advanced **machine learning models** to estimate the performance ratio of solar PV plants using operational and environmental data. Among tested algorithms, **Random Search CV Regression** achieved the highest accuracy with an **R² of 0.93** and minimal error. Results highlight the effectiveness of models like **Random Search CV**, **AdaBoost**, and **Random Forest** in optimizing solar plant performance and advancing efficient renewable energy solutions.

A Comparison Between Machine Learning Models for Air ticket Price Prediction

IEEE
📅 11/2022 🔗 <https://ieeexplore.ieee.org/document/9998230>
Explored **machine learning techniques** to predict **airline ticket prices** using two Kaggle datasets, addressing challenges from dynamic pricing fluctuations. Evaluated models including **Random Forest** and **Randomized Search CV**, with the proposed model providing accurate fare predictions. This solution supports travelers in making informed decisions by forecasting price trends from historical data.

CERTIFICATION

AWS Certified Solutions Architect – Associate Amazon Web Services (AWS)	SQL for Data Science University of California, Davis
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LANGUAGES

English Proficient		German Intermediate	
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